B. Tech. Degree III Semester Examination November 2013

IT/ME 1302 ELECTRICAL TECHNOLOGY

(2012 Scheme)

Time : 3 Hours | Maximum Marks : 100

PART A
(Answer *ALL* questions)

(8 x 5 = 40)

I. (a) Derive the emf equation of a transformer.

(b) What do you mean by CT and PT? Where are they used?

(c) Explain the conditions for voltage build up of a self excited DC shunt generator.

(d) Explain the significance of back emf in a dc motor.

(e) Explain how the torque is developed in a rotor of 3ϕ induction motor.

(f) Explain the different methods of starting synchronous motors.

(g) List the various electrical equipments used in power stations.

(h) Discuss the advantage of high voltage transmission.

PART B

(4 x 15 = 60)

II. The following readings were obtained from OC and SC tests on 3KVA, 220V/110V, 50Hz transformer. (15)
OC test (lv side : 110V, 0.9A, 38W
SC test (hv side) : 15V, 13.6A, 45W

Calculate:

(i) the equivalent circuit constants w.r.t. HV side
(ii) voltage regulation and efficiency for 0.8 lagging power factor load.
(iii) The efficiency at half full load and 0.8 pf load.

OR

III. (a) Draw and explain the no load phasor diagram of a single phase transformer (5)

(b) A transformer rated for 10KVA, 2000V/200V, 50Hz when tested took 200W on OC test at rated voltage and 250W on SC test with full load current circulated in the windings. Determine the efficiency at full load, 0.8 pf lagging and the KVA loading to give maximum efficiency at upf. (10)

IV. (a) Derive the emf equation of a generator (5)

(b) A shunt motor takes an armature current of 50A at 250V when running on full load, at a speed of 800 rpm. The armature resistance is $0.2\,\Omega$. If the field strength is reduced by 10% and the torque remains the same, determine the steady speed attained and the armature current. (10)

OR

V.	(a)	Explain the effect of armature reaction. How is it compensated?	(7)
	(b)	A dc shunt generator is having 600 wave connected conductors. The flux per pole is 0.06wb and the machine is having 6 poles. The armature is rotating at a speed of 900rpm. The armature resistance is $0.05\,\Omega$, field circuit resistance is $100\,\Omega$. Calculate the output of the generator if the terminal voltage is 1550V.	(8)
VI.	(a)	Define voltage regulation of an alternator. Explain the EMF method of finding voltage regulation.	(7)
	(b)	A three phase 16 pole alternator has a star connected winding with 144 slots and 10 conductors/slot. The flux/pole is 0.03 wb sinusoidally distributed and the speed is 375 rpm, find the frequency, the phase and line emf. Assume full pitched coil.	(8)
		OR	
VII.		A 6 pole 50Hz 3ϕ induction motor running on full load with 3% slip develops a torque of 160Nm at its pulley rim. The friction and windage losses are 210W and stator copper and iron losses equal to 1640W. Calculate. (i) Rotor slip (ii) Efficiency at Full load (iii) Rotor Cu loss	(15)
VIII.		Explain the working of thermal power plant with the help of neat schematic diagram	(15)
		OR	
IX.	(a)	What is corona? What are the factors affecting corona?	(7)
	(b)	Explain hydroelectric power generation with neat sketch.	(8)

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